C Programming A Modern Approach

Chapter 3 Formatted Input/Output[1]

1. Write a program that accepts a date from the user in the form *mm/dd/yyyy* and then displays it in the form *yyyymmdd*:

   ```
   Enter a date (mm/dd/yyyy):2/17/2011
   You entered the date 20110217
   ```

   Answer: Source Code

2. Write a program that formats product information entered by the user. A session with the program should look like this:

   ```
   Enter item number: 583
   Enter unit price: 13.5
   Enter purchase date (mm/dd/yyyy): 10/24/2010

   Item            Unit            Purchase
                   Price           Date
   583             $  13.50        10/24/2010
   ```

   The item number and date should be left justified: the unit price should be right justified. Allow dollar amounts up to $9999.99. *Hint:* Use tabs to line up the columns.

   Answer: Source Code

3. Books are identified by an International Standard Book Number (ISBN). ISBNs assigned after January 1, 2007 contain 13 digits, arranged in five groups, such as 978-0-393-97950-3. (Older ISBNs use 10 digits.) The first group (the *GSI prefix*) is currently either 978 or 979, The *group identifier* specifies the language or country of origin (for example, 0 and 1 are used in English-speaking countries). The *publisher code* identifies the publisher to (393 is the code for W. W. Norton). The *item number* is assigned by the publisher to identify a specific book (97950 is the code for this book). An ISBN ends with a *check digit* that's used to verify the accuracy of the preceding digits. Write a program that breaks down an ISBN entered by the user:

   ```
   Enter ISBN: 978-0-393-97950-3
   GSI prefix: 978
   Group identifier: 0
   Publisher code: 393
   Item number: 97950
   Check digit: 3
   ```

   Note: The number of digits in each group may vary; you can't assume that groups have the lengths shown in this example. Test your program with actual ISBN values (usually found on the back cover of a book and on the copyright page).

   Answer: Source Code

4. Write a program that prompts the user to enter a telephone number in the form (xxx)xxx-xxxx and then displays the number in the form xxx.xxxx.xxx:

   ```
   Enter phone number [(xxx) xxx-xxxx]: (404) 817-6900
   You entered 404.817.6900
   ```

   Answer: Source Code

5. Write a program that asks the user to enter the numbers from 1 to 16 (in any order) and then displays the numbers in a 4 by 4 arrangement, followed by the sums of the rows, columns, and diagonals:

   ```
   Enter the numbers from 1 to 16 in any order:
   16 3 2 13 5 10 11 8 9 6 7 12 4 15 14 1

   16  3  2 13
    5 10 11  8
    9  6  7 12
    4 15 14  1

   Row sums: 34 34 34 34
   Column sums: 34 34 34 34
   Diagonal sums: 34 34
   ```

   If the row, column, and diagonal sums are all the same (as they are in this example), the numbers are said to form a ***magic square***. The magic square shown here appears in a 1514 engraving by artist and mathematician Albrecht Dürer. (Note that the middle numbers in the last row give the date of the engraving.)

   Answer: Source Code

6. Modify the `addfrac.c` program of Section 3.2 so that the user enters both fractions at the same time, separated by a plus sign:

   ```
   Enter two fractions separated by a plus sign: 5/6+3/4
   The sum is 38/24
   ```

   Answer: Source Code

Bibliography

[1] K. N. King, *C Programming A Modern Approach*. 2008, pp. 50–51.